The effects of a Porous Layer on the two-way radar attenuation on Enceladus

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Introduction

Saturn's moon, Enceladus is considered a priority target for future planetary missions due to its high astrobiological potential [1]. Water jets presumably originating from a subsurface ocean have been observed at the south pole of Enceladus by NASA's Cassini mission [2], and their analysis provides a direct window into the ocean composition [3] that, in turn, can help to understand the nature and amount of impurities that may exist within the ice shell.

Enceladus' jet activity generates a highly porous material that affects the thermal state of the ice shell. The thickness of that layer and its distribution are poorly constrained, but local thicknesses of up to 700m have been reported from the analysis of pit chains on the surface of Enceladus [4]. Such a thick porous layer can strongly attenuate the signal of radar sounders that have been proposed to investigate the Enceladus' subsurface [5].

Method – 2-way Radar Attenuation

Electrical conductivity (including impurities)

$$\sigma = \sum_{i=0}^{2} \sigma_i^0 C_i \exp\left[-\frac{E_i}{k_B}\left(\frac{1}{T}-\frac{1}{T_r}\right)\right]$$

1-way radar attenuation

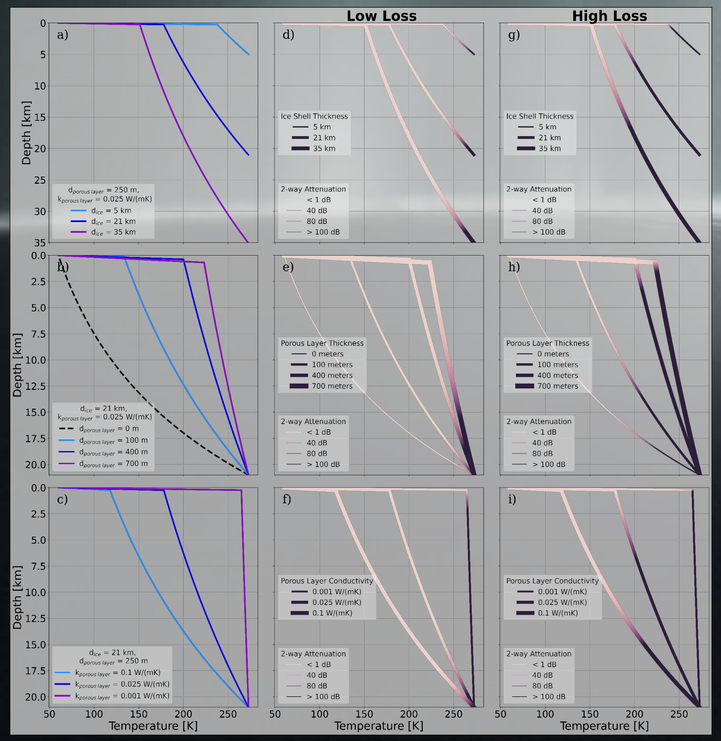
$$A_1 = \frac{1000(10\log_{10} e)}{c\,\varepsilon_0\sqrt{\varepsilon_r}}\sigma \approx 0.914\,\sigma$$

2-way radar attenuation

$$A_2(d) = \sum_{i=0}^{i_d} 2A_1(d_i)\Delta d$$

Results – Effects of Porous Layer Properties

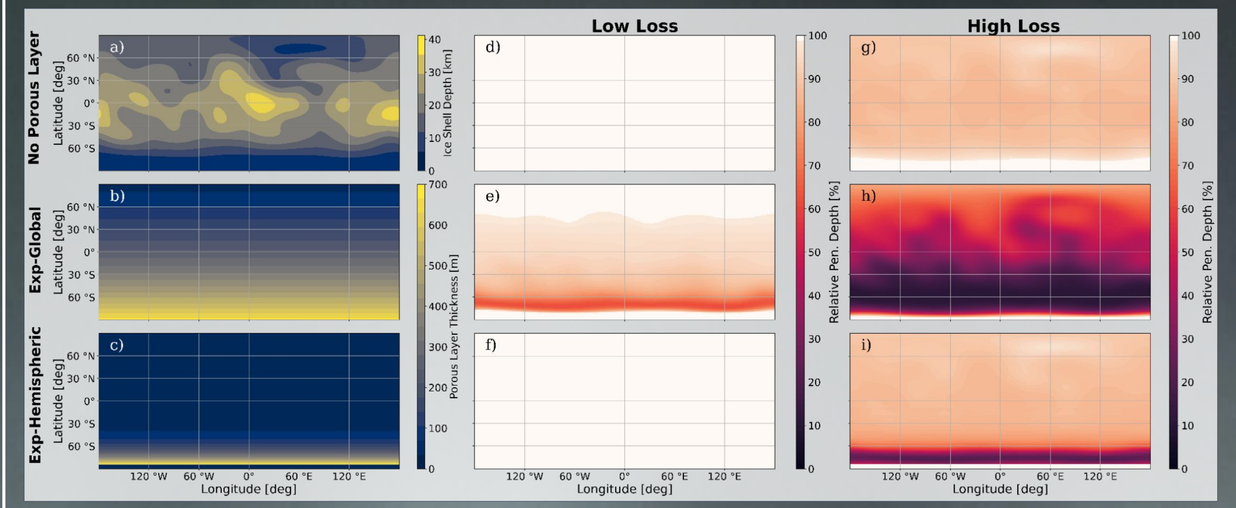
- The thicker the porous layer is, the higher the subsurface temperature.
- The lower thermal conductivity of the porous layer is, the higher the subsurface temperature.



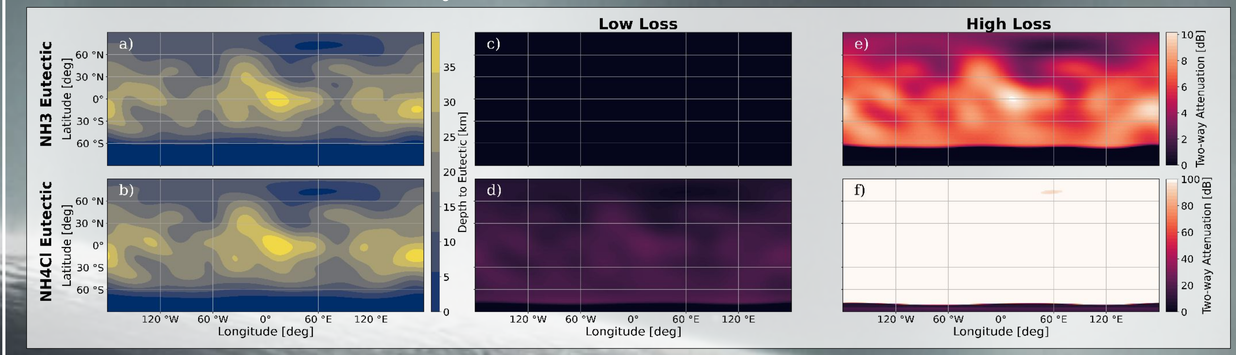


Results – Effects of Porous Layer Distribution

- Ice shell thickness map from Hemingway and Mittal, (2019) [6]
- Porous layer distribution assuming exponential decay from south to north (panel d)
- Porous layer distribution assuming deposition only on the southern hemisphere (panel g)



- High ice shell temperatures: formation of shallow brines that could provide a radar-detectable source
- Eutectic interface of ammonium chloride (NH_4Cl): 257.79 K isotherm
- Eutectic interface of ammonia (NH_3): 175.45 K isotherm



Take Home Messages

- The porous layer thickness and its distribution have a first order effect on the 2-way radar attenuation in Enceladus' ice shell
- Regions covered by a thick porous layer may prevent the radar signal from reaching the ice-ocean interface.
- However, the high temperatures may lead to the formation of shallow brines that would be detected by the radar measurements.
- Such brine reservoirs are considered niches for life, and their potential to be directly accessible by future missions is much greater compared to that of the oceans.

References: [1] Choblet et al., 2021. [2] Hansen et al., 2006. [3] Postberg et al., 2008. [4] Martin et al., 2023. [5] Soucek et al., 2023. [6] Hemingway & Mittal, 2019.